Inventory Optimization and the Economic Order Quantity

A Guided-Discovery Approach with Interactive Analytics

Chenhao Zhou

Rutgers Business School

Department of Supply Chain Management

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Today's Roadmap

Part I: The Inventory Problem ~8 min

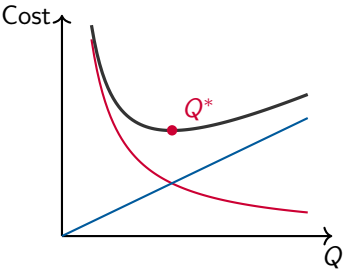
Real-life scenario, industry context, case setup.

Part II: Interactive Discovery ~12 min

Colab notebook: build formulas, plot curves, discover the optimal point *by doing*.

Part III: Why & What's Next ~5 min

Mathematical hint, homework, preview.

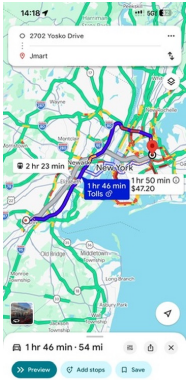


Part I

The Inventory Problem

Personal scenario $\rightarrow$ Industry context $\rightarrow$ Case study $\rightarrow$ Model

A Personal Inventory Problem



Edison, NJ → JMart, Flushing

54 mi · 1h 46m · Tolls $47.20

I drive to **JMart in Flushing** to stock up on frozen dumplings, bao, sauces, noodles—daily-use items stored in a chest freezer.

Every round trip costs me:

- Tolls + Gas: $47 + $18 = **$65 fixed per trip**

Storing at home costs me:

- Extra chest freezer electricity: ∼$12/month
- Food quality degrades over time

My dilemma: Buy a lot each trip (fewer $65 trips, but more freezer cost) or go more often?

The Same Dilemma—At Scale

Every Supply Chain Faces This

A restaurant ordering ingredients, a hospital restocking medical supplies, a retailer replenishing shelves—all must decide: **how much to order, and how often?**

Two costs pull in **opposite directions**:

Ordering Cost

Fixed cost per order—paperwork, shipping, receiving, inspection.

Order *large* $Q \Rightarrow$ fewer orders $\Rightarrow$ **lower** ordering cost

Holding (Carrying) Cost

Cost of keeping inventory—space, capital, spoilage, insurance.

Order *large* $Q \Rightarrow$ more inventory $\Rightarrow$ **higher** carrying cost

There must be a sweet spot in between.

How big is this problem in the real economy?

How Big Is This Problem?

Source: CSCMP / Kearney, *Annual State of Logistics Report*.

Year	Inv. Value ($B)	Carry Rate	Carry Cost ($B)	% of Logistics
2010	2,064	19.2%	396	32.7%
2014	2,495	19.1%	476	32.8%
2018	2,583	22.1%	571	34.8%
2022	∼2,700	∼31%	∼852	∼31.9%
2024	**∼2,580**	**∼28%**	**∼735**	**28.5%**

US inventory carrying costs ≈ **$735 billion/year**.
Roughly **one in four dollars** spent on logistics goes to simply *holding* inventory.

Where the Money Goes—By Industry

	Food & Grocery	Electronics / Tech	Pharma / Medical	Auto Parts
Capital / interest	6–10%	10–15%	8–12%	8–12%
Warehousing	3–5%	2–4%	4–8%	2–4%
Spoilage / obsolescence	**8–15%**	**10–20%**	3–6%	3–5%
Insurance / taxes	1–2%	1–3%	2–4%	1–2%
Total carry rate ≈	20–30%	25–40%	18–28%	15–22%
Ordering: key driver	Delivery freq.	Supplier MOQ	Compliance	Setup time

- **Food/grocery**: spoilage dominates—freshness decays daily
- **Electronics**: obsolescence is huge—product cycles of 6–12 months
- **Pharma**: regulatory storage costs (cold chain, controlled substances)
- **Auto parts**: long product life, but capital tied up in slow-moving SKUs

Case Study: San Antonio's Specialty Foods

Your Assignment

San Antonio's is a local specialty food distributor. They sell artisanal pasta, imported olive oil, and gourmet sauces to restaurants across the region.

Your boss walks in and says:

"We order from our supplier every week, and our warehouse is always either overflowing or running low. I need your team to figure out: ***how many cases should we order each time to minimize our total inventory cost?*** *Build me a model and show me the numbers."*

What we know:

- Customer demand is stable: $\sim$1,000 cases/month = **12,000 cases/year**
- Product prices are stable (long-term supplier contract)
- We need to manage *supply costs*: ordering and holding

The Model: Setting Up Parameters

Inventory Cost Model

Parameters

D	$= 12{,}000$	Annual demand (cases/yr)
B	$= \$60$	Fixed ordering cost ($/order)
H	$= \$16$	Holding cost ($/case/yr)

Decision Variable

Q	$= ?$	Order quantity (cases/order)

Objective: $$\min_{Q} \; TC(Q) = \underbrace{\frac{DB}{Q}}_{\text{ordering}} + \underbrace{\frac{QH}{2}}_{\text{carrying}}$$

Assumptions: Constant demand, entire order arrives at once, no stockouts, zero/constant lead time.

Part II

Interactive Discovery with Colab

Build the model $\rightarrow$ Plot each curve $\rightarrow$ Discover Q^*

Colab Prompt 1: Set Parameters

Open Google Colab. Paste this prompt into the AI assistant or a new cell:

Prompt 1

"Set up an inventory cost model for San Antonio's Specialty Foods.
Parameters: D = 12,000 cases/year, B = $60/order, H = $16/case/year. Create a range of Q from 50 to 800.
Print the parameters to confirm."

Think of this as the header of our optimization model—like declaring constants in a math program.
Everything that follows will reference these three numbers: D, B, H.

Ordering Cost: $\frac{DB}{Q}$

Formula:

$$\text{Ordering cost} = \frac{D}{Q} \times B = \frac{DB}{Q}$$

Number of orders/yr $= D/Q$.
Each order costs B.
Total: DB/Q per year.

Q (cases)	Ordering ($/yr)
100	$7,200
200	$3,600
300	**$2,400**
600	$1,200

As Q increases ↑: cost decreases ↓
(Fewer trips to the supplier)

Prompt 2

"Plot the **ordering cost** DB/Q as a function of Q (50 to 800). Use a red line. Label axes: 'Order Quantity Q' and 'Annual Cost ($)'. Set y-axis 0 to 12,000."

You should see a **decreasing curve** (hyperbola).

Intuition: Larger orders → fewer orders per year → lower total ordering cost.

Carrying Cost: $\frac{QH}{2}$

Formula:

$$\text{Carrying cost} = \frac{Q}{2} \times H = \frac{QH}{2}$$

Average inventory $= Q/2$.
Each unit costs H per year to hold.
Total: $QH/2$ per year.

Q (cases)	Carrying ($/yr)
100	$800
200	$1,600
300	**$2,400**
600	$4,800

As Q increases ↑: cost increases ↑
(More inventory sitting in the warehouse)

Prompt 2 (continued)

"Now add the **carrying cost** $QH/2$ to the same chart as a blue line. Show both ordering (red) and carrying (blue) on the same plot."

You should see an **increasing line** crossing the red curve.

Intuition: Larger orders → more inventory on hand → higher storage cost.

Quiz: What Should You Do?

Quiz 1: When the **holding cost** H **is very large** (expensive warehouse, perishable goods), should you order in **large** quantities or **small** quantities?

A) Large Q, infrequently
B) Small Q, frequently

Quiz 2: When the **ordering cost** B **is very large** (expensive shipping, long setup), should you order in **large** quantities or **small** quantities?

A) Large Q, infrequently
B) Small Q, frequently

Colab Prompt 3: Total Cost—All Three Curves

Prompt 3

"Now compute the **total cost** $TC(Q) = DB/Q + QH/2$.
Plot all three curves on the same chart: ordering (red), carrying (blue), total (black, thicker).
Calculate $Q^* = \sqrt{2DB/H}$ and mark it with a red star on the chart.
Draw a dashed vertical line at Q^*.
Title the chart with the Q^* value and the minimum total cost.
Print the ordering cost and carrying cost at Q^* separately."

Look at the chart carefully.

Where does the total cost curve hit its minimum?

What happens at that exact point on the red and blue curves?

What Do You Notice?

The output from Prompt 3 should show:

At $Q^* = 300$:

$$\begin{aligned} \text{Ordering cost} &= \$2{,}400 \\ \text{Carrying cost} &= \$2{,}400 \\ \hline \end{aligned}$$

They are exactly equal!

Is this a coincidence for our numbers ($D = 12{,}000$, $B = \$60$, $H = \$16$)?
Or is it *always* true that Q^* sits where the two curves **cross**?

Your Turn: Try Different Parameters

Exercise: Go back to Cell 1 and change the parameters to:

```
D = 8000,    B = 100,    H = 20
```

Re-run **all cells** (Cells 1–5). Then answer:

1. What is the new Q^*?
2. What is the new TC^*?
3. At Q^*, is ordering cost still equal to carrying cost?

Your Turn: Try Different Parameters

Exercise: Go back to Cell 1 and change the parameters to:

```
D = 8000,   B = 100,   H = 20
```

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1. What is the new Q^*?
2. What is the new TC^*?
3. At Q^*, is ordering cost still equal to carrying cost?

Try a few more sets of numbers on your own.

Does the pattern always hold?

Live Demo: It Always Holds

Colab is powerful, but what if we want a **drag-and-explore** tool?

We prompt Claude (or ChatGPT) directly:

```
"Create an interactive HTML page with an EOQ chart.
Plot ordering cost DB/Q in red, carrying cost QH/2 in blue,
total cost in dark.  Add sliders for D, B, H.
Fix the axes (x:  0-800, y:  0-12000) so only the curves move.
Mark Q* with a star."
```

The AI generates a complete, self-contained HTML file.
Open it in any browser. **Drag the sliders**:

- The minimum *always* sits at the intersection
- Ordering cost always equals carrying cost at Q^*
- This is not a coincidence—it is a **mathematical certainty**

Let's see the live demo...

Part III

Why? & What's Next

A mathematical hint for homework

Homework: Why Does This Always Work?

We observed: *no matter what D, B, H we choose, the minimum of $TC = \frac{DB}{Q} + \frac{QH}{2}$ always occurs where $\frac{DB}{Q} = \frac{QH}{2}$.*

Hint: Start from here

For any real numbers a and b:

$$(a-b)^2 \geq 0$$

Expand this. Rearrange. What do you get?

Homework details on the next slide.

Homework (continued)

Your task (due next class):

1. Expand $(a-b)^2 \geq 0$ and show that $a^2 + b^2 \geq 2ab$.
2. Let $a = \sqrt{\frac{DB}{Q}}$ and $b = \sqrt{\frac{QH}{2}}$. What happens?
3. When does equality hold? What does that mean for Q?
4. Solve for Q^*. Does your formula match $Q^* = \sqrt{\frac{2DB}{H}}$?

We will prove this together next class using the **AM–GM inequality**.

Summary & What's Next

Today

- The EOQ balances **ordering cost** (DB/Q) and **carrying cost** ($QH/2$)
- We discovered Q^* visually using Colab and an interactive tool
- Key insight: at Q^*, ordering cost = carrying cost—**always**

Upcoming: (1) EOQ proof via AM–GM (2) Quantity discounts (3) Demand forecasting (4) Reorder points & safety stock

Thank You & Questions

Chenhao Zhou

School of Business, Rutgers Camden · Supply Chain Management

chenhao.zhou@rutgers.edu · chenhaozhou.me